

Can the environmental impact of ChatGPT and other AI be measured?

Posted by u/Responsible_Roof6171 • 0 points • 7 comments

As the title asks.

We know that AI uses energy and has a negative impact on the environment, but are we able to quantify this, and ideally compare it to other things, such as the meat industry / car manufacturing etc?

Comments

u/Lumpy-Notice8945 • 1 points • 2025-08-03 03:20 UTC

Ofc it can be measured, its the cost to run the computer the AI is running and was trained on.

Training is a one time cost and its quite big, running the AI is not that much, it runs on nvidia GPUs that use about as much power as a modern gaming PC, so like 800-1200W and it takes chatGPT a couple of secon6to generate a reply.

u/Advanced-List-4483 • 0 points • 2025-08-03 03:18 UTC

CAN we? 100% yes, we can.

ARE we doing it? Well... At least in the US, the parts of the EPA that usually run the modeling for these large-scale calculations are currently in the process of being gutted by the Trump administration. So I would suspect that any 'official' statistics on the environmental impacts of AI are going to be quietly buried.

Nevertheless, there are nonprofits and other organizations that also make these kinds of estimates. But since the companies that run these large datacenters are unlikely to want to share exact numbers (for obvious reasons) those will be estimates.

u/archpaw • 2 points • 2025-08-03 03:18 UTC

[It's really not that much.](<https://andymasley.substack.com/p/a-cheat-sheet-for-conversations-about>)

One prompt is believed to use about 3 Wh of electricity (probably less). [Here's a list of other things that use that much.]

(<https://docs.google.com/document/d/1pDdpPq3MyPdEAoTkho9YABZ0NBEhBH2v4EA98fm3pXQ/edit?tab=t.0>) For example, using a toaster for nine seconds is as bad as prompting ChatGPT.

u/Advanced-List-4483 • 1 points • 2025-08-03 03:22 UTC

Sure, but you don't use your toaster for only nine seconds, do you? And you don't ask ChatGPT JUST ONE prompt.

The question is what is the total, cumulative impact of *all* users.

u/MshaCarmona • 0 points • 2025-08-03 02:58 UTC

Literally the best way to get this answer is chatpgt. Any chatgpt question us best answered by chatgpt

u/Concise_Pirate • 3 points • 2025-08-03 02:57 UTC

Yes, absolutely. The companies that run these services know exactly how much electricity, and how much cooling water, their computers are using. They pay the power bill and the water bill every month.

u/inolyzushi • 2 points • 2025-08-03 03:02 UTC

Add to that, the minerals being mined for GPUs and chips - I would think that the process is being escalated, accelerated more than ever.